

Anti-Carjack System



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Consider this: A criminal stops your car by force and you are left with no option but to surrender the car. The criminal drives away your car, but about 60 seconds later the engine stops running on its own and won't restart. Unable to use the dead vehicle, the perpetrator has to abandon it.

Carjacking can be prevented using the anti-carjack system described here. The circuit automatically senses carjacking and stops the vehicle. In case the circuit is accidentally tripped off, it can be easily reset using a hidden switch. Two LEDs indicate the status of the system to the vehicle driver.

Circuit description

Fig. 1 shows the circuit of the anti-carjack system. It comprises quad NAND gate CD4011 (IC1), timer NE555 (IC2), optocoupler MCT2E (IC3) and a few discrete components. The circuit is powered by 12V DC from the vehicle battery. Diode D1 protects against any possible reverse voltage. The main controlling elements of the circuit are NAND gates N1 and N2 and timer NE555.

The input to gate N2 comes from a car accessory power line (such as cigarette lighter power socket) that is energised by the ignition switch. The input to gate N1 comes from the door switch that connects the dome light. The door switches in the vehicle provide the ground return for the dome or courtesy light when any of the doors is opened.

The line connecting the door switch to the lamp is at 12V when the door is closed and at zero when any door is open. Logic signal from the dome light is fed through diode D2 to pins 12 and 13 of gate N1, which inverts it to drive pin 2 of gate N2. At the same time, the accessory power line, which is at 12V when ignition

switch is turned on, connects to input pin 1 of gate N2 through diode D3.

Under normal conditions, when the ignition switch is 'on' and the doors closed, the output of gate N2 at pin 3 is high. However, when a door is opened, the output of gate N2 at pin 3 goes low. This negative-going output from gate N2 is coupled to the trigger input (pin 2) of NE555 (IC2), which is configured as a monostable multivibrator. Output pin 3 of IC2 goes high for a period determined by the combination of resistor R4 and capacitor C5.

The two remaining NAND gates (N3 and N4) of CD4011 are connected in a bistable multivibrator configuration with two input terminals, pins 8 and 6. The bistable configuration has two stable states with logic outputs at pin 4 and pin 10 always being opposite to each other. A negative transition at the input terminal of either gate N3 or N4 causes the bistable to change the state.

PARTS LIST

Semiconductors:	
IC1	- CD4011 quad NAND gate
IC2	- NE555 timer
IC3	- MCT2E optocoupler
T1	- BS170 MOSFET
T2	- BC337 npn transistor
D1-D3, D5	- 1N4007 rectifier diode
D4	- 1N4148 switching diode
LED1	- 5mm green LED
LED2	- 5mm red LED
Resistors (all 1/4-watt, $\pm 5\%$ per cent carbon):	
R1, R2	- 4.7-kilo-ohm
R3, R5, R6	- 10-kilo-ohm
R4	- 1-mega-ohm
R7, R8	- 470-ohm
R9	- 680-ohm
R10	- 2-kilo-ohm
Capacitors:	
C1	- 100 μ F, 35V electrolytic
C2, C3	- 4.7 μ F, 35V electrolytic
C4, C6, C7	- 0.1 μ F ceramic disk
C5	- 47 μ F, 35V electrolytic
C8	- 0.01 μ F ceramic disk
Miscellaneous:	
RL1	- 12V, 1C/O relay
S1	- Push-to-on tactile switch
	- Ignition coil

When the time period of NE555 (IC2) is over, its output pin 3 goes low. This negative-going signal is coupled to the input of gate N4 causing the bistable circuit to change the state. As a result, pin 4 of gate N4 goes high.

The high output of gate N4 forward-biases the internal LED of optocoupler MCT2E (IC3), driving the internal transistor into saturation. As a result, relay RL1 energises and its normally-closed (N/C) contact opens to cut the power to the ignition coil and stall the vehicle. When reset switch S1 is pressed, a negative transition is applied to pin 8 of gate N3. Pin 4 of gate N4 goes low to de-energise the relay and allow the vehicle to start.

The green LED (LED1) driven by MOSFET T1 glows to alert the vehicle owner if the timer has been activated, either by a carjack attempt or by inadvertent opening of a door when the ignition switch is 'on'. LED1 gives the driver a warning that the circuit has been triggered and a lock-out will occur in about 60 seconds. The red LED (LED2) glows whenever relay RL1 energises to stall the vehicle.

The circuit is designed to be 'fail safe'. The N/C contact of RL1 is used to control vehicle operation. The relay doesn't energise unless the circuit is powered and the timer triggered.

Any component failure may cause the relay to energise. To de-energise it, open jumper JP to cut off power to the relay coil circuit so that RL1 contacts remain closed.

Construction and testing

An actual-size, single-side PCB for the anti-carjack system is shown in Fig. 3 and its component layout in Fig. 4. Assemble the circuit on a PCB as it minimises time and assembly errors. Carefully assemble the components and double-check for any overlooked error. Before inserting the ICs in the PCB, properly check the supply voltage at each IC's supply terminals. Suitable connectors are provided on the PCB for connections to accessory, dome light and battery supply. Use suitable